

Performance Work Statement (PWS)
National Oceanic and Atmospheric Administration (NOAA)
Oceanic and Atmospheric Research (OAR)
Great Lakes Environmental Research Laboratory (GLERL)

**3rd Generation (3G) Environmental Sample Processor (ESP) Surface Plasmon Resonance (SPR)
Integration and Support Services**

1. INTRODUCTION

1.1. Mission

GLERL is responsible for advancing the scientific understanding and ability to detect, monitor, assess, and predict harmful algal bloom (HAB) in direct support of 33 U.S.C. 53.

1.2. Background

HABs are expanding globally, threatening economic development, fisheries, and human and ecosystem health. The increase in HABs is exceeding our monitoring capacity to track bloom events and subsequent ecosystem impacts. Biogeochemical processes in shallow, coastal systems and remote areas, where ecosystem dynamics are changing rapidly, cannot be observed using traditional observing platforms alone. In the Laurentian Great Lakes cultural eutrophication within freshwater ecosystems has been linked to the increased presence of cyanobacteria HABs (cHABs). In certain regions of the Great Lakes ecosystem, cHABs have been a recurrent summer feature since the mid-1990s. The most predominant cHAB is *Microcystis aeruginosa*, which produces expansive blooms in western Lake Erie and Lake Huron (Saginaw Bay). While remote regions such as the Alaskan Arctic are experiencing accelerated warming and emerging HABs, mainly *Alexandrium* spp., have impacted the region's human and ecosystem health and multi-billion-dollar fishing industry.

1.3. Scope

This PWS covers the use and assistance in deploying, embedding into an uncrewed surface vehicle (USV), and calibration of a 3G ESP with an SPR analytical module, which is designed to collect both eDNA samples and determine phycotoxin concentrations in situ. The project includes assessing the USV capability to conduct event-based response deployments, monitoring harmful algal blooms, understanding phytoplankton community dynamics, and assisting with public health associated with phycotoxins in Great Lakes and Alaska. USVs are a rapidly evolving technology that can provide real-time continuous reporting to shore, integrate sensors or sampling devices at multiple water depths, while permitting sampling operations in shallow and remote systems. To better inform management actions and public health, near real time information is needed on the multiple interactive factors that influence HAB formation, duration and phycotoxin levels.

2. GENERAL REQUIREMENTS

2.1. Non-Personal Services

The Government shall neither supervise contractor employees nor control the method by which the contractor performs the required tasks. Under no circumstances shall the Government assign tasks

to, or prepare work schedules for, individual contractor employees. It shall be the responsibility of the contractor to manage its employees and to guard against any actions that may be deemed as personal services.

2.2. Contract Administration and Management

The Contractor shall establish processes and assign appropriate resources to effectively administer the requirement. The Contractor shall respond to Government requests for contractual actions in a timely fashion. The Contractor shall have a single point of contact between the Government and contractor personnel assigned to support contracts or task orders. The Contractor shall assign work effort and maintain proper and accurate time keeping records of personnel assigned to work on the requirement.

2.3. Project Management

The contractor shall attend coordination and project meetings, assist with manuscript and presentation development, and provide oversight throughout the performance of this project, ensuring that Monterey Bay Aquarium Research Institute (MBARI) is involved in all project planning and meetings, and is part of the onsite crew for all missions.

2.4. Location of Work

Work required by this PWS will be performed at GLERL in Ann Arbor, MI during performance supporting the Great Lakes mission and at Nome, AK during performance supporting the Alaska mission.

2.5. Travel

Travel to other government facilities or other contractor facilities may be required. All travel requirements (including plans, agenda, itinerary, or dates) shall be pre-approved by the Government (subject to local policy & procedures) and will be reimbursed on a time-and-materials (T&M) basis. Costs for travel shall be billed in accordance with the regulatory implementation of Public Law 99-234 and FAR 31.205-46 Travel Costs (subject to local policy & procedures; may reference FAR).

3. PERFORMANCE REQUIREMENTS

3.1. Water Capture Sampling System Integration Services

3.1.1. The contractor shall design, fabricate, and deliver a water capture sampling system. The sample capture system needs to have the ability to collect at least 1000 ml of water, in which, the 3G ESP-SPR system is able to sample from and move the water sample inside a USV platform. The water capture system must include a pressure gauge and venting system to ensure bubbles are removed prior to 3G ESP-SPR sampling, once sampling is completed the water capture system needs to drain any remaining captured water

3.1.2. The contractor shall install a water capture sampling system for the 3G ESP into the SeaTrac platform. The water sample aggregation system for the 3G ESP system must be fully embedded into the USV platform no later than June 2026.

3.2. 3G ESP-SPR Hardware Configuration and Payload Integration Services

3.2.1. The contractor shall deliver the 3G ESP-SPR system and auxiliary parts.

3.2.2. The contractor shall build and deliver cartridges that are properly built to prevent fluid bypass, ensuring that all cartridges and cartridge parts are clean. Cartridges must contain a 5 µm Durapore membrane filter. Depending on cartridge type, Archive or Lyse-n-go, the sample preservation or processing reagents must be included in the cartridge. The 3G ESPs pump system must be validated through measuring the volume dispensed by the rotating piston pump and verify the cartridge carousel is indexed correctly for cartridge type (Archive or Lyse-n-go) are properly seated to prevent fluid bypass (operating pressure between 20 - 40 psi) to ensure the system does not leak.

3.2.3. The contractor shall integrate SPR system into the 3G ESP, which includes SPR calibrations through known target injection and standard curve generation that is suitable for environmental conditions and phycotoxin to be analyzed prior to full integration into the 3G ESP system.

3.2.4. The contractor shall complete full installation of the 3G ESP-SPR system, which includes placing 3G ESP-SPR with cooling system into pressurized housing, pressurize housing and testing for air leaks, securing straps on the 3G ESP-SPR housing, and insulating 3G ESP-SPR system with double reflective insulation. The 3G ESP-SPR system must be fully functional and calibrated, ensuring the 3G ESP-SPR system can be embedded into the SeaTrac platform, no later than 24 hours prior to the start of all GLERL field missions.

3.2.5. The contractor shall provide all up-to-date SPR operation coding and provide standard curve data, including providing code responsible for all 3G ESP-SPR operations and training, as needed, or other hardware designs. The operational command codes to run 3G ESP-SPR system must be provided no later than 24 hours prior to the start of all GLERL field missions.

3.3. Mission Support Services

3.3.1. The contractor shall assist with coding or error issue troubleshooting and solutions from the water capture system and 3G ESP-SPR system during all mission deployments.

3.3.2. The contractor shall assist in recovery of samples at the end of the mission. This includes completing positive and negative checks of Archive or Lyse-n-go samples and removal of cartridges based on indexed map filter removal.

3.3.3. The contractor shall recover the 3G ESP-SPR system and provide post operation calibration of the 3G ESP-SPR system.

3.3.4. The contractor shall ensure the 3G ESP-SPR system is returned to MBARI upon mission completion.

3.3.5. The contractor shall produce a final data set with comprehensive metadata, in coordination with GLERL, upon completion of all GLERL field missions. NOAA will submit data to the NOAA National Centers for Environmental Information (NCEI), which will provide public access and permanent archiving, within a year of project completion. The archive link will be included in all relevant peer-reviewed publications. This final data set will also be on a NOAA public website (https://www.glerl.noaa.gov/res/HABs_and_Hypoxia/habsMon.html).

4. Deliverables

PWS Ref.	Deliverable	Due Date	Medium	Submit to
3.1.1.	Water capture sampling system	June 2026	Physical Delivery	Contracting Officer Representative (COR)
3.2.1.	3G ESP-SPR system and auxiliary parts	No later than 24 hours prior to field mission	Physical Delivery	COR
3.2.2.	Cartridges	No later than 24 hours prior to field mission	Physical Delivery	COR
3.2.5.	All up-to-date SPR operation coding and standard curve data	No later than 24 hours prior to field mission	Spreadsheet and standard curve graphics	COR
3.3.3	Post deployment QC checks	Upon field mission completion	Spreadsheet and standard curve graphics	COR
3.3.5.	Final data set with comprehensive metadata	Upon field mission completion	Spreadsheet and standard curve graphics	COR

5. Performance Requirement Summary (PRS)

PWS Ref.	Performance Standard	Quality Level	Method of Surveillance	Incentive/Disincentive
3.1.2.	Fully embedded water capture sampling system for the 3G ESP system into the USV platform successfully no later than June 2026	100%	Inspection and acceptance	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS
3.2.3.	Successfully completed full installation of the fully functional and calibrated 3G ESP-SPR system no later than 24 hours prior to the start of GLERL field missions.	100%	Inspection and acceptance	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS
3.3.1.	Successfully resolve or provide documented workarounds for coding errors and hardware-software interface within an hour of detection during active mission deployment of the water capture sampling or 3G ESP-SPR systems, ensuring zero loss of critical sample data.	100%	Inspection and acceptance	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS

3.3.2.	Successfully recovers samples, completing positive and negative checks of samples, and cartridge removal at the end of GLERL field missions.	100%	Inspection and acceptance	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS
3.3.3.	Successfully recover the 3G ESP-SPR system and provide post operation calibration of the 3G ESP-SPR system at the end of GLERL field missions.	100%	Inspection and acceptance	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS
3.3.4.	Successfully return the 3G ESP-SPR system to MBARI upon completion of field missions	100%	Confirmation with MBARI	Incentive – Favorable CPARS Disincentive -Corrective action plan submitted to COR for review; unfavorable CPARS